

Trial Test 3

- Submit your answers via Google Forms (you have only 1 hour and 10 mins to complete).
- Your score will be available immediately after the test. Please, check your emails.
- If you have technical issues, let me know in DM or in the Group Chat.
- You are not allowed to use the calculator. Also, do not google, be honest!
- Do not panic and manage your time wisely. Read the questions carefully and try to answer all of them.

Good Luck!

1) Evaluate:

$$(\sqrt{7} + \sqrt{3})^2 - (\sqrt{7} - \sqrt{3})^2$$

a) $2\sqrt{7}$

b) $4\sqrt{7}$

c) $2\sqrt{21}$

d) $4\sqrt{21}$

2) Find the complete set of values of x which satisfy the inequality:

$$\frac{1}{2}(3x - 2) - \frac{2}{3}(x - 4) < x$$

a) $x > 10$

b) $x < 22$

c) $x > 2.5$

d) $x < 4.5$

3) The equation of a curve is $y = px^2 + qx$ where p and q are constants.

The curve passes through the points $(2, 6)$ and $(4, -4)$.

What is the value of $q - p$?

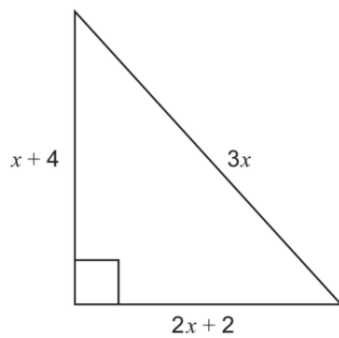
a) 16

b) 9

c) 5

d) 1

4) The diagram shows a right-angled triangle, with sides of length $x+4$, $2x+2$ and $3x$, all in cm. Find the area.



- a) 54
- b) 70
- c) 36
- d) 40

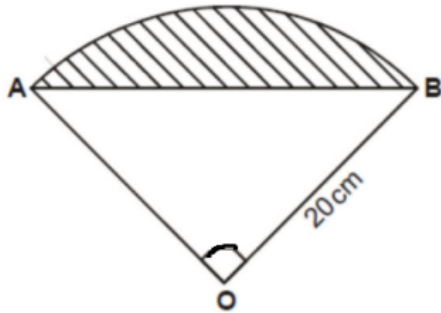
5) Express as a single fraction:

$$\frac{1}{2x} + \frac{1}{x-1} - \frac{1}{x}$$

- a) $\frac{1}{2x-1}$
- b) $\frac{2x-1}{x(x-1)}$
- c) $\frac{x+1}{2x(x-1)}$
- d) $\frac{x-3}{2x(x-1)}$

6) The diagram shows the sector of circle, centre O, radius 20 cm and angle AOB is 150 degrees.

Find the area of the segment and take π as 3.



- a) 300
- b) 250
- c) 200
- d) 400

7) A metal alloy is 25% copper. Another metal alloy is 50% copper.

How much of 25% alloy should be used to make 500 g of an alloy that is 45% copper?

- a) 200
- b) 300
- c) 100
- d) 150

8) Two similar cones have radii of 4 cm and 6 cm. The total area of smaller one is

$$72\pi \text{ cm}^2.$$

Find the total area of the larger cone.

a) 162π

b) 184π

c) 216π

d) 98π

9) The equation connects the variables M, x, y, z, P and Q . x and y are both increased by 50%, z is decreased by 20%, P is doubled, Q remains the same.

What is the percentage change of M ?

$$M = \frac{(x+y)^2 z}{P} Q$$

a) 20% increase

b) 20 decrease

c) 10% increase

d) 10% decrease

10) One side of a rectangle is $(7 - \sqrt{5})$ cm. The rectangle has an area of 66 cm^2 .

What is the perimeter, in cm, of the rectangle?

a) $\frac{35-5\sqrt{5}}{2}$

b) $35 - 5\sqrt{5}$

c) $\frac{35+5\sqrt{5}}{2}$

d) $35+\sqrt{5}$

11) p is directly proportional to $(q + 2)^2$. When $q=1$, $p=1$.

Find p when $q=10$.

a) 22

b) 16

c) 14

d) 20

12) The design in the diagram is formed from four circles which all touch at the top of the shape (the diagram is not to scale). The diameter of the smallest circle is d , the second is $2d$, the third is $3d$ and the largest is $4d$.

Find an expression for the area of the shading in the design.



a) $\frac{5}{2}\pi d^2$

b) $4\pi d^2$

c) $\frac{3}{2}\pi d^2$

d) $10\pi d^2$

13) What is the coefficient of the x^2 term in the binomial expansion of $(2 - x)^5$?

- a) -48
- b) 40
- c) -88
- d) 80

14) Simplify

$$3x\left(\frac{3x^7}{x^{\frac{1}{3}}}\right)^3$$

- a) $9x^{20}$
- b) $27x^{20}$
- c) $3x^{21}$
- d) $81x^{21}$

15) Write 72^2 as a product of its prime factors.

- a) $2^6 * 3^4$
- b) $2^4 * 3^4$
- c) $2^2 * 3^6$
- d) $2^{4*} 3^2$

16) The first five terms of a sequence in order are:

2 17 42 77 122

The n^{th} term of this sequence is $pn^2 + q$ where p and q are integers.

What is the value of $\frac{p-q}{p+q}$?

- a) 14
- b) 4
- c) 2
- d) $\frac{23}{17}$

17) The total of three numbers p , q and r is 375

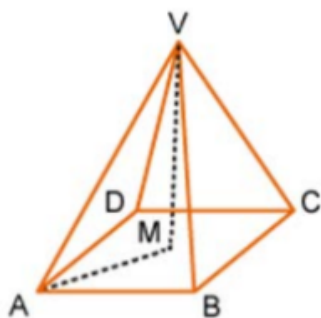
The ratio $p:q$ is 5:7. The ratio $q:r$ is 4:11.

What is the value of $p+r$?

- a) 315
- b) 231
- c) 291
- d) 144

18) In the figure below, the square base ABCD has side lengths of $\sqrt{20}$ cm. Lengths VA, VB, VC and VD are each 8 cm. M is the midpoint of the square.

What is the volume of this pyramid?



- a) $24\sqrt{6}$

b) $32\sqrt{3}$

c) $20\sqrt{6}$

d) $48\sqrt{3}$

19) Five apples, three pears and two bananas cost 3.18\$. Four apples, eight pears and three bananas cost 4.49\$.

How much more expensive is an apple than a pear?

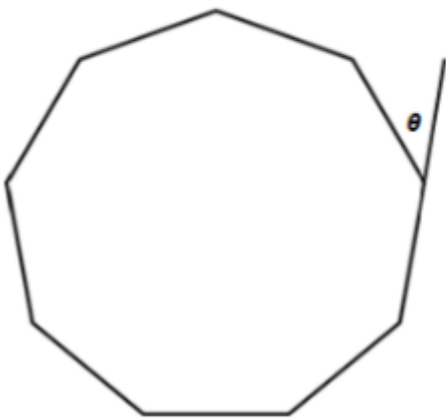
a) $10p$

b) $6p$

c) $7p$

d) $8p$

20) In the regular polygon shown, find the measure of angle θ .



a) 45 deg

b) 40 deg

c) 50 deg

d) 30 deg

21) The roots of the equation $2x^2 - 11x + a = 0$ differ by 2. The value of a is:

a) $\frac{105}{8}$

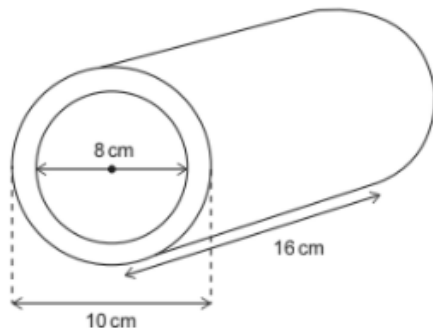
b) $\frac{113}{8}$

c) $\frac{117}{8}$

d) $\frac{119}{8}$

22) A cylindrical hollow metal pipe is 16 cm long. It has an external diameter of 10 cm, and an internal diameter of 8 cm. The density of the metal from which the pipe is made is 8 grams per cm^3 .

What is the mass of the pipe in grams?



a) 512π

b) 1024π

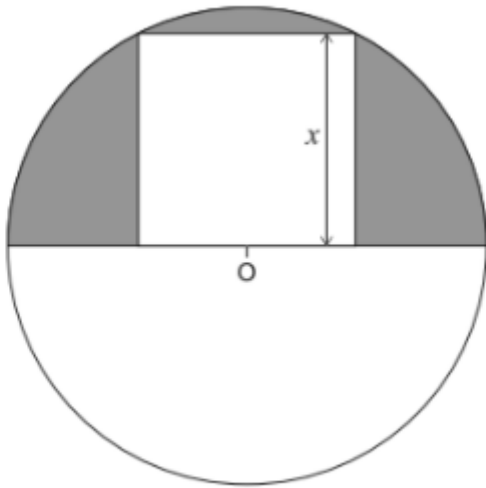
c) 256π

d) 1152π

23) The diagram shows a square with a side of length x cm. A circle is drawn with centre O which lies at the midpoint of one of the sides of the square. The side forms

part of a diameter of the circle. The circle passes through two corners of the square as shown.

What is the area, in cm^2 , of the shaded part of the semi-circle?



a) $\left(\frac{5\pi-8}{8}\right)x^2$

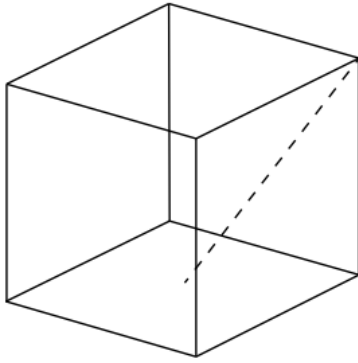
b) $\left(\frac{5\pi-4}{4}\right)x^2$

c) $\left(\frac{3\pi-2}{2}\right)x^2$

d) $\left(\frac{3\pi-4}{4}\right)x^2$

24) A cube has sides of unit length.

What is the length of a line joining a vertex to the midpoint of one of the opposite faces (the dashed line in the diagram below)?



a) $\sqrt{\frac{3}{2}}$

b) $\sqrt{2}$

c) $\sqrt{\frac{5}{2}}$

d) $\sqrt{3}$

25) If you look at a clock at 9:45, what is the angle between the hour and minute hands?

a) 30 deg

b) 15 deg

c) 7.5 deg

d) 22.5 deg

26) Which of the expressions below has the largest value for $0 < x < 1$?

a) $\frac{1}{x}$

b) x^2

c) $\frac{1}{(1+x)}$

d) $\frac{1}{\sqrt{x}}$

27) Find the sum of the solutions of

$$2\left(\frac{x}{4} + 3\right)^2 - \left(\frac{x}{4} + 3\right) - 36 = 0$$

a) -22

b) -34

c) -36

d) -13

28) Calculate $(1.63 * 10^{12}) * (2.47 * 10^{-1})$.

Give your answer in standard form.

a) $402.61 * 10^9$

b) $4.0261 * 10^{11}$

c) $0.40261 * 10^{12}$

d) $40.261 * 10^{10}$

29) The weight of an animal increases by 50% each year for 3 years. It then declines in weight by 20% a year for 2 years, and then declines by a further 25% in the next year.

What is the percentage increase in the animal's weight at the end of 6 years compared to its original weight at the start of this period?

a) 85%

b) 20%

c) 62%

d) 115%

30) S and T are geometric progressions. For each, the second term is 6 and the sum to infinity is 25. The first term of S is greater than the first term of T.

What is the fourth term of S?

a) $\frac{8}{125}$

b) $\frac{4}{25}$

c) $\frac{24}{25}$

d) $\frac{162}{125}$